SIMATS SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI-602105
TECHNICAL QUESTIONS FOR HBS – SET2

Engineer to Excel

TECHNICAL QUESTIONS ROUND 1

1. Consider these two functions and two statements S1 and S2 about them.

```
int work1(int *a, int i, int j)
{
int x = a[i+2];
a[j] = x+1;
return a[i+2] - 3;
}
int work2(int *a, int i, int j)
{
int t1 = i+2;
int t2 = a[t1];
a[j] = t2+1;
return t2–3;
}
```

S1: The transformation from work1 to work2 is valid, i.e., for any program state and input arguments, work2 will compute the same output and have the same effect on program state as work1

S2: All the transformations applied to work1 to get work2 will always improve the performance (i.e reduce CPU time) of work2 compared to work1

(a) S1 is false and S2 is false
(b) S1 is false and S2 is true
(c) S1 is true and S2 is false
(d) S1 is true and S2 is true

Answer: (d)

2. Consider the following C function:

```
int f(int n)
{
static int r = 0;
if (n <= 0) return 1;
if (n > 3)
{
r = n;
return f(n-2)+2;
}
return f(n-1)+r;
}
```

What is the value of f(5) ?

(a) 5 (b) 7 (c)9 (d) 18

Answer: (d)

3. What is the return value of the function foo when it is called as foo(345, 10) ?

(a) 345 (b) 12 (c) 5 (d) 3

Answer: (b)

4. What is the return value of the function foo when it is called as foo(513, 2)?

(a) 9 (b) 8 (c) 5 (d) 2

Answer: (d)

5. Choose the correct to fill ?1 and ?2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a newline character.

```
void reverse(void)
{
int c;
if (?_1) reverse() ;
?_2
}
main()
{
printf ("Enter Text ") ; printf ("\n") ;
reverse(); printf ("\n") ;
}
```

(a) ?1 is (getchar() != '\n'') **and** ?2 is getchar(c);
(b) ?1 is (c = getchar()) != '\n') **and** ?2 is getchar(c);
(c) ?1 is (c != '\n') **and** ?2 is putchar(c);
(d) ?1 is ((c = getchar()) != '\n') **and** ?2 is putchar(c);

Answer: (d)

6. The following C declaration

```
struct node
{
int i;
float j;
};
struct node *s[10] ;
```

define s to be

(a) An array, each element of which is a pointer to a structure of type node
(b) A structure of 2 fields, each field being a pointer to an array of 10 elements
(c) A structure of 3 fields: an integer, a float, and an array of 10 elements
(d) An array, each element of which is a structure of type node.

Answer: (a)

7. The most appropriate matching for the following pairs

X: m=malloc(5); m= NULL; 1: using dangling pointers
Y: free(n); n->value=5; 2: using uninitialized pointers
Z: char *p; *p = 'a'; 3: lost memory

is:

(a) X — 1 Y — 3 Z-2
(b) X — 2 Y — 1 Z-3
(c) X — 3 Y — 2 Z-1
(d) X — 3 Y — 1 Z-2

Answer: (d)

8. Consider the following C declaration

```
struct {
short s [5]
union {
float y;
long z;
}u;
} t;
```

Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t, ignoring alignment considerations, is

(a) 22 bytes (b) 14 bytes (c)18 bytes (d) 10 bytes

Answer: (c)

9. Assume the following C variable declaration

```
int *A [10], B[10][10];
```

Of the following expressions

I. A[2]
II. A[2][3]
III. B[1]
IV. B[2][3]

which will not give compile-time errors if used as left hand sides of assignment statements in a C program?

(a) I, II, and IV only
(b) II, III, and IV only
(c) II and IV only
(d) IV only

Answer: (a)

10. Consider the following C function

```
void swap (int a, int b)
{
int temp;
temp = a;
a = b;
b = temp;
}
```

In order to exchange the values of two variables x and y.

a) call swap (x, y)
b) call swap (&x, &y)
c) swap (x, y) cannot be used as it does not return any value
d) swap (x, y) cannot be used as the parameters are passed by value

Answer: (d)

11. What does the following C-statement declare?

```
int ( * f) (int * ) ;
```

(a) A function that takes an integer pointer as argument and returns an integer
(b) A function that takes an integer as argument and returns an integer pointer
(c) A pointer to a function that takes an integer pointer as argument and returns an integer.
(d) A function that takes an integer pointer as argument and returns a function pointer

Answer: (c)

12. What does the following program print?

```
#include<stdio.h>
void f(int *p, int *q)
{
p = q;
*p = 2;
}
int i = 0, j = 1;
int main()
{
f(&i, &j);
printf("%d %d \n", i, j);
getchar();
return 0;
}
```

(a) 2 2 (b) 2 1 (c)0 1 (d) 0 2

Answer: (d)

13. What does the following fragment of C-program print?

```
char c[] = "GATE2011";
char *p =c;
printf("%s", p + p[3] - p[1]);
```

(a) GATE2011 **(b)** E2011 **(c)** 2011 **(d)** 011

Answer: (c)

14. Consider the following C program segment:

```
char p[20];
char *s = "string";
int length = strlen(s);
int i;
for (i = 0; i < length; i++)
p[i] = s[length - i];
printf("%s",p);
```

The output of the program is

(a) gnirts
(b) gnirt
(c) string
(d) no output is printed

Answer: (d)

15. What output will be generated by the given code segment?

(a) || (b) || © || (d)
3 1 || 4 2 || 4 2 || 3 1
4 1 || 6 1 || 6 2 || 5 2
4 2 || 6 1 || 2 0 || 5 2

Answer: (c)

16. What output will be generated by the given code segment if:
Line 1 is replaced by “auto int a = 1;
Line 2 is replaced by “register int a = 2;

(a) || (b) || (c) || (d)
3 1 || 4 2 || 4 2 || 4 2
4 1 || 6 1 || 6 2 || 4 2
4 2 || 6 1 || 2 0 || 2 0

Answer: (d)

17. What is printed by the following C program?

```
int f(int x, int *py, int **ppz)
{
int y, z;
**ppz += 1;
z = **ppz;
*py += 2;
y = *py;
x += 3;
return x + y + z;
}
void main()
{
int c, *b, **a;
c = 4;
b = &c;
a = &b;
printf( "%d", f(c,b,a));
getchar();
}
```

(a) 18 (b) 19 (c) 21 (d) 22

Answer: (b)

18. The output of the following C program is _____.

```
#include <stdio.h>
void f1(int a, int b)
{
int c;
c=a; a=b; b=c;
}
void f2(int *a, int *b)
{
int c;
c=*a; *a=*b;*b=c;
}
int main()
{
int a=4, b=5, c=6;
f1 (a, b);
f2 (&b, &c);
printf("%d", c-a-b);
}
```

Answer: -5

19. What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.

```
int main()
{
```

```
unsigned int x[4][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}, {10, 11, 12}};
printf("%u,%u, %u", x+3, *(x+3),*(x+2)+3);
}
```

(a) 2036, 2036, 2036
(b) 2012, 4, 2204
(c) 2036, 10, 10
(d) 2012, 4, 6

Answer: (a)

20. Consider the following C function.

```
int fun1 (int n)
{
int i, j, k, p, q = 0;
for (i = 1; i<n; ++i)
{
p = 0;
for (j=n; j>1; j=j/2)
++p;
for (k=1; k<p; k=k*2)
++q;
}
return q;
}
```

Which one of the following most closely approximates the return value of the function fun1?

(a) n^3
(b) $n(\log n)^2$
(c) n log n
(d) n log(log n)

Answer: (d)

ADVANCED CODING QUESTIONS ROUND 2

1. Given an unlimited supply of coins of given denominations, find the minimum number of coins required to get the desired change.
   For example, consider S = { 1, 3, 5, 7 }.

   If the desired change is 15, the minimum number of coins required is 3 (7 + 7 + 1) or (5 + 5 + 5) or (3 + 5 + 7)
   If the desired change is 18, the minimum number of coins required is 4 (7 + 7 + 3 + 1) or (5 + 5 + 5 + 3) or (7 + 5 + 5 + 1)

2. Word Break Problem: Given a string and a dictionary of words, determine if the string can be segmented into a space-separated sequence of one or more dictionary words.

   Input:
   dict[] = { this, th, is, famous, Word, break, b, r, e, a, k, br, bre, brea, ak, problem };
   word = Wordbreakproblem
   Output:
   Word b r e a k problem

Word b r e ak problem
Word br e a k problem
Word br e ak problem
Word bre a k problem
Word bre ak problem
Word brea k problem
Word break problem

3. Given a rod of length n and a list of rod prices of length i, where 1 <= i <= n, find the optimal way to cut the rod into smaller rods to maximize profit.
   For example, consider the following rod lengths and values:
   Input:
   length[] = [1, 2, 3, 4, 5, 6, 7, 8]
   price[] = [1, 5, 8, 9, 10, 17, 17, 20]
   Rod length: 4
   Best: Cut the rod into two pieces of length 2 each to gain revenue of 5 + 5 = 10

Cut	**Profit**
4	9
1, 3	(1 + 8) = 9
2, 2	(5 + 5) = 10
3, 1	(8 + 1) = 9
1, 1, 2	(1 + 1 + 5) = 7
1, 2, 1	(1 + 5 + 1) = 7
2, 1, 1	(5 + 1 + 1) = 7
1, 1, 1, 1	(1 + 1 + 1 + 1) = 4

4. Given an array arr[] where each element represents the max number of steps that can be made forward from that index. The task is to find the minimum number of jumps to reach the end of the array starting from index 0.
   Examples:
   Input: arr[] = {1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9}
   Output: 3 (1-> 3 -> 9 -> 9)
   Explanation: Jump from 1st element to 2nd element as there is only 1 step. Now there are three options 5, 8 or 9. If 8 or 9 is chosen then the end node 9 can be reached. So 3 jumps are made.

   Input: arr[] = {1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1}
   Output: 10
   Explanation: In every step a jump is needed so the count of jumps is 10.